

Homophilic Networks and Demographic Winter: A Threshold Model for the Impact of Pension System Stress.



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CONCLUSIONS

- The targeted exit of "Young Graduates" (network Hubs) fragments the labor market's relational capital.
- Austerity policies act as a negative feedback loop, directly accelerating brain drain.
- Structural network erosion far exceeds linear demographic loss (a 5.8% drop in nodes destroys 31.4% of network links).

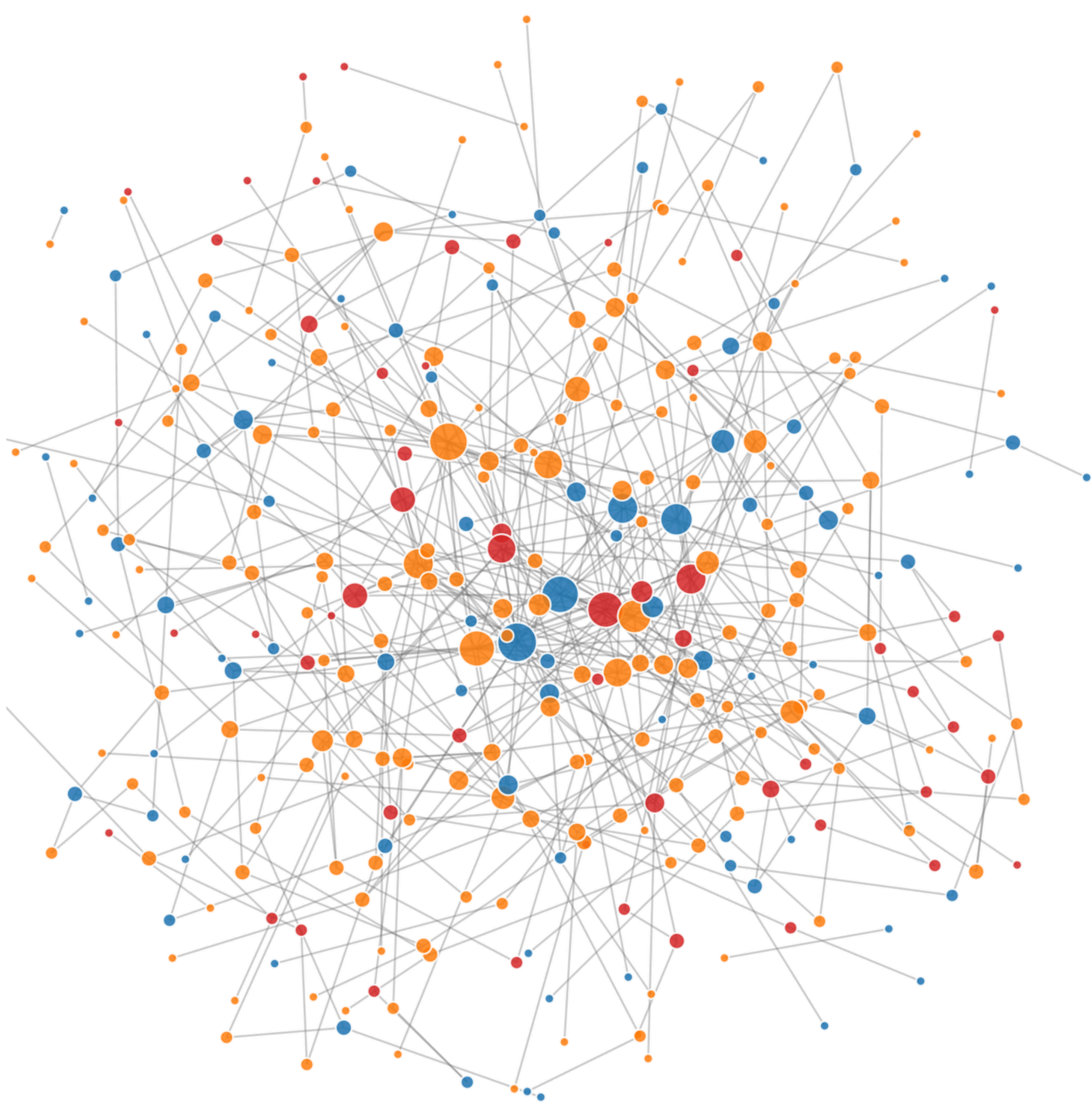
INTRODUCTION & AIM

- Italy faces a severe demographic winter and an unsustainable pay-as-you-go pension system.
- Traditional macro-models treat the labor market as a homogeneous aggregate, failing to capture social contagion and individual behavior.
- Objective: Simulate how homophilic network topology amplifies or attenuates emigration cascades following a fiscal shock.

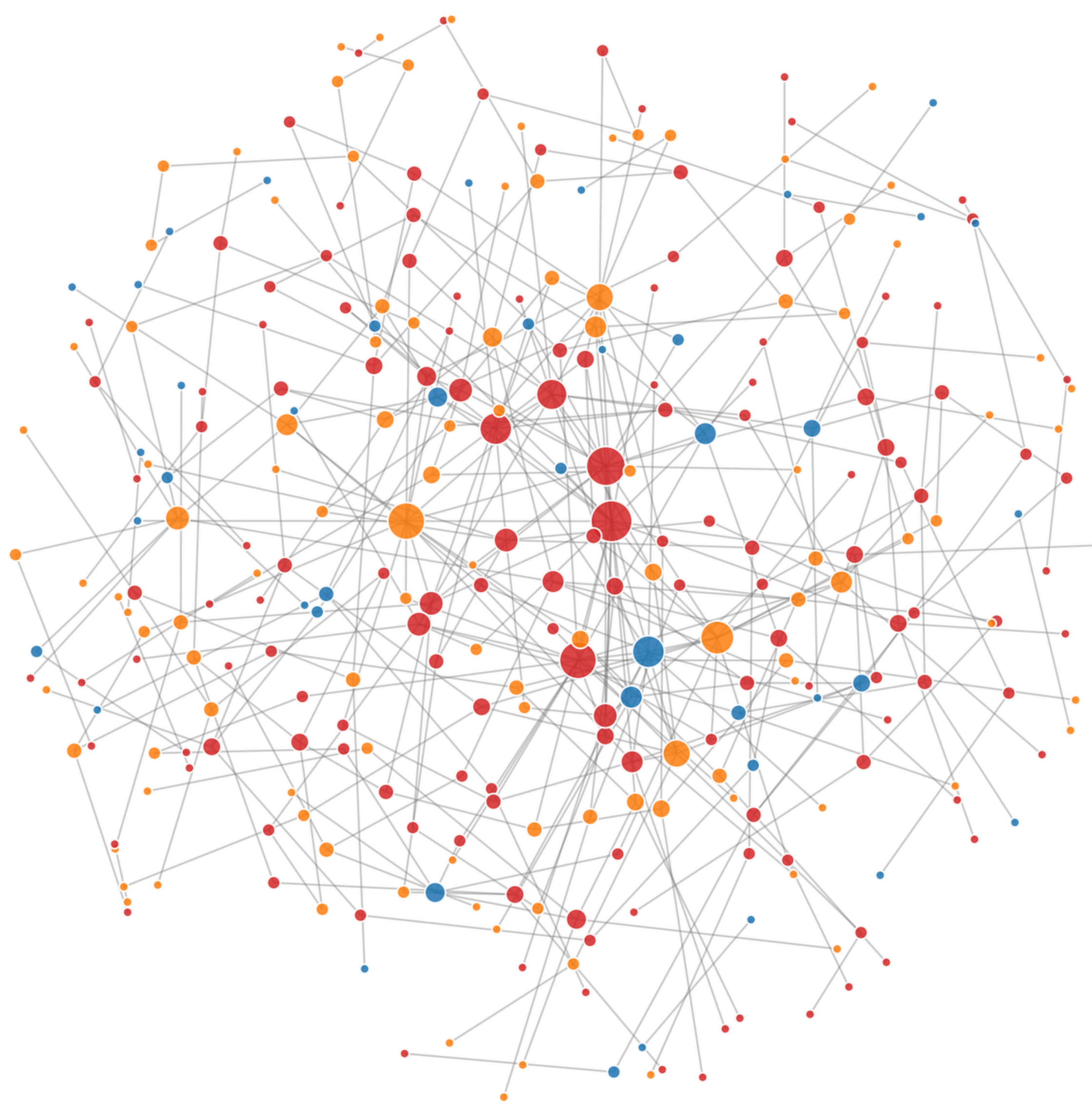
METHODS

- Network Generation: A 5,000-node synthetic Scale-Free network using the Barabási-Albert algorithm, calibrated on ISTAT data.
- Behavioral Dynamics: Granovetter's threshold model regulates individual decisions to emigrate or enter the informal economy.
- Evolutionary Engine: A 20-year cycle tracking demographic turnover, tax evasion, and systemic fiscal stress.

Initial Topology (Year 0)



Final Topology: System Fragmentation (Year 20)



● Young Graduates (Brain Drain) ● Adults (Fiscal Backbone) ● Elderly (Demographic Winter)

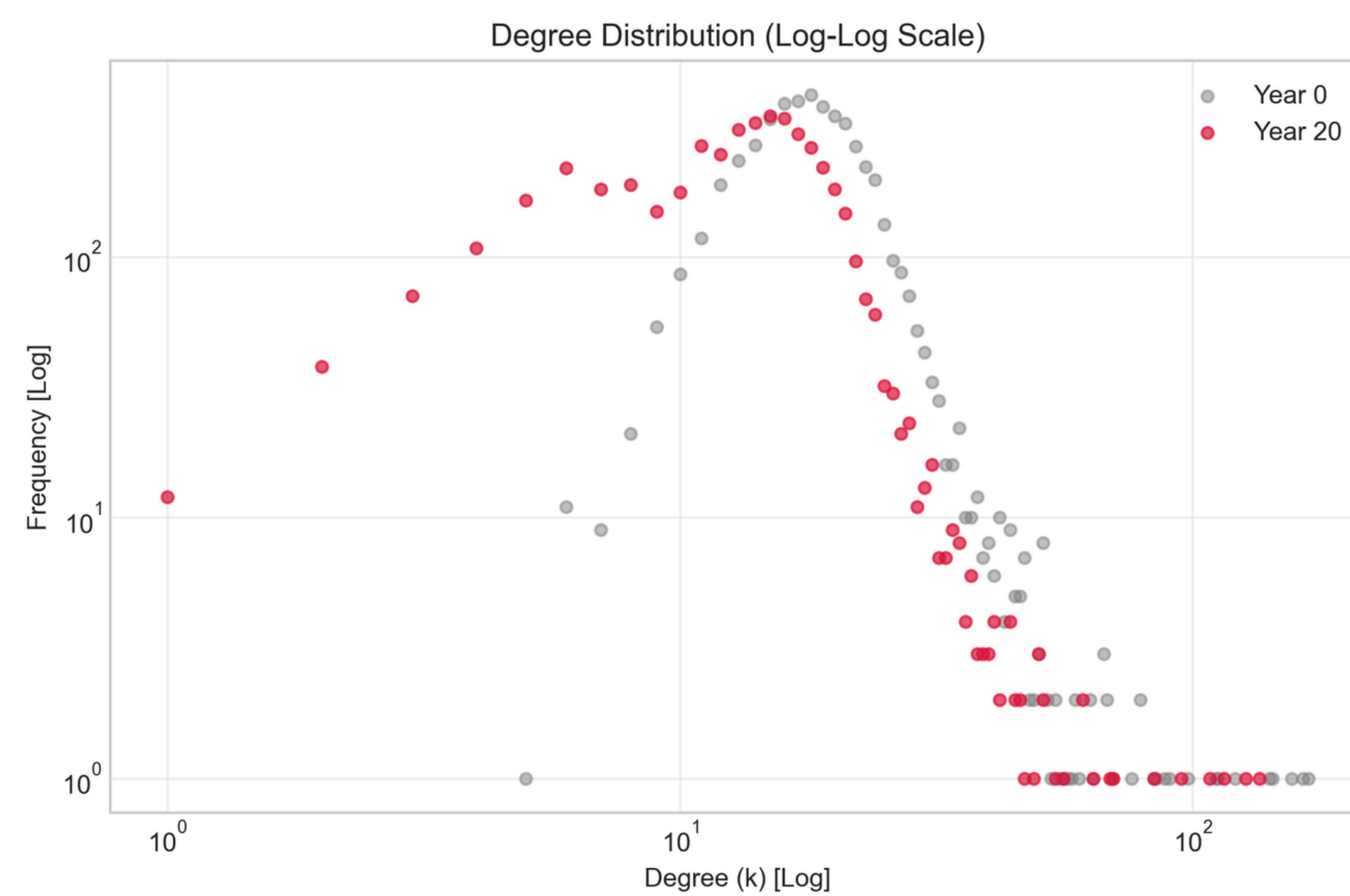
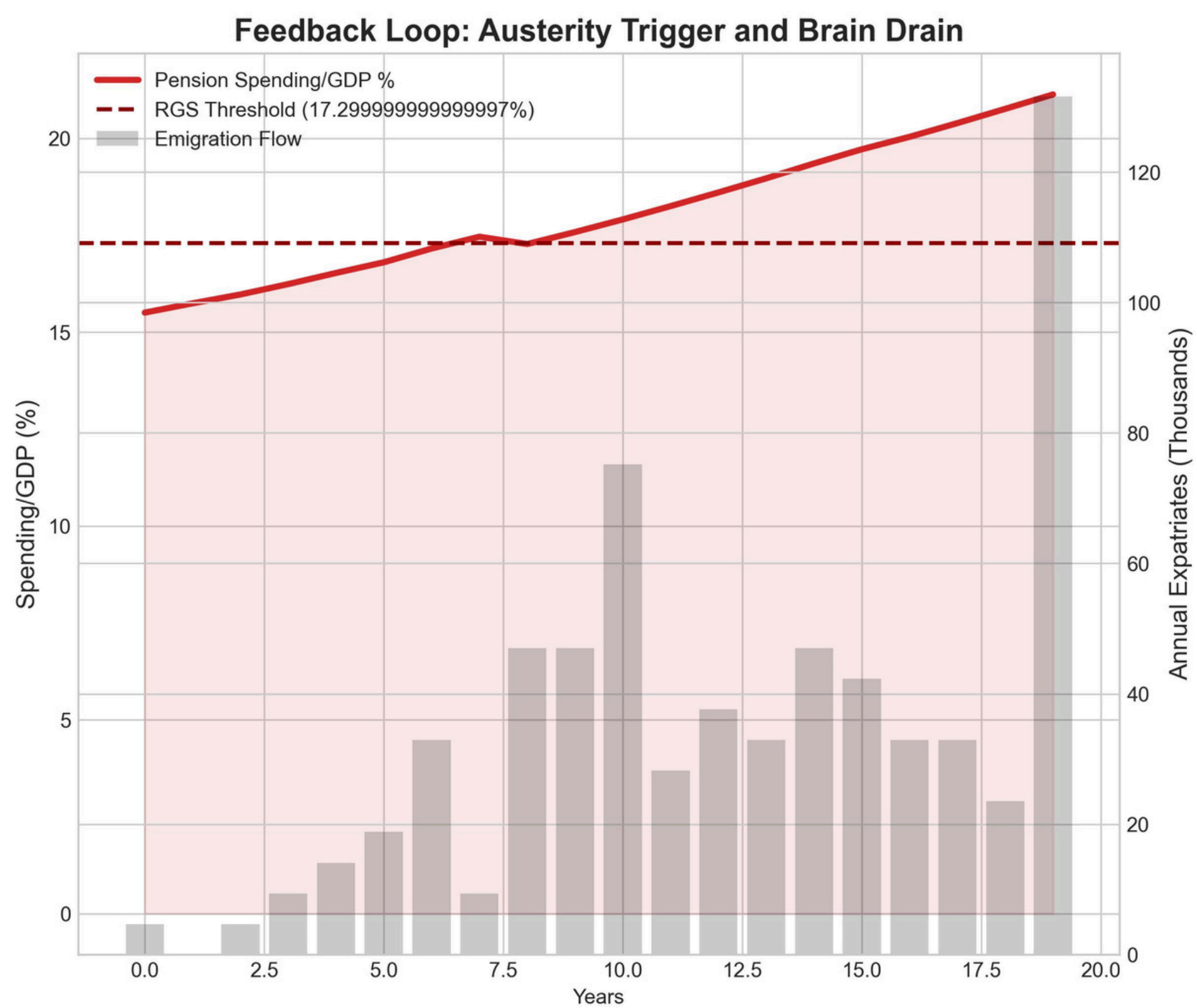
DATASET & PYTHON CODE

Scan or click to access the synthetic Scale-Free network data and Granovetter's simulation script.



RESULTS

- Austerity Trigger: When pension spending crosses the 17.3% GDP threshold, fiscal correction lowers systemic resilience, driving non-linear cascades of espatriation.
- Topological Collapse: Brain drain destroys the network's "long tails," forcing the economic system to rely on older, less-connected nodes.



REFERENCES

- Barabási, A.-L., & Albert, R. (1999). Science, 286(5439).
- Granovetter, M. (1978). American Journal of Sociology, 83(6).
- MEF - Ragioneria Generale dello Stato (2025). Report n.26.